

# Schur Apolarity and the Dual Terracini Lemma for Flag Varieties: A Geometric and Representation-Theoretic Approach

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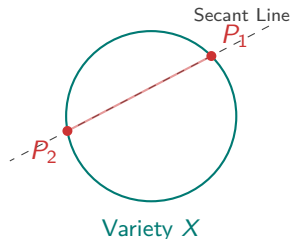
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# The Problem: Secant Varieties and Tensors

- **Secant Varieties:** Classical objects in algebraic geometry, bridging projective geometry and tensor decomposition.
- **Definition:** Let  $X \subset \mathbb{P}^N$ . The  $s$ -th secant variety  $\sigma_s(X)$  is

$$\sigma_s(X) = \overline{\bigcup_{P_1, \dots, P_s \in X} \langle P_1, \dots, P_s \rangle}.$$



## Terracini's Lemma

The tangent space to the secant variety at a generic point is the span of the tangent spaces at the generic points  $P_i \in X$ :

$$T_Q \sigma_s(X) = \langle T_{P_1} X, \dots, T_{P_s} X \rangle.$$

- **Defectivity:** We define  $\exp.\dim(\sigma_s(X)) := \min\{N, s(\dim X + 1) - 1\}$ . If  $\dim(\sigma_s(X)) < \exp.\dim(\sigma_s(X))$ , then  $X$  is said to be *defective*.

# Motivation: Classical Apolarity and Lasker's Result

For **Veronese varieties**  $X_{n,d} \subset \mathbb{P}(\text{Sym}^d V)$  (symmetric tensors), the problem is well-understood. A key tool is a result attributed to Emanuel Lasker.

## Lasker's Lemma (Veronese Case)

The classical apolar space to the tangent space at a point  $P = [v^d] \in X_{n,d}$  is the degree- $d$  component of the square of the ideal of  $p = [v]$ :

$$(\widehat{T}_P X_{n,d})^\perp = I^2(p)_d.$$

**The Dual Terracini Lemma.** By dualizing Terracini's Lemma, we convert the geometric span into an algebraic intersection:

$$(\widehat{T}_{Q\sigma_r(X)})^\perp = \bigcap_{i=1}^r (\widehat{T}_{P_i} X)^\perp = \bigcap_{i=1}^r I^2(p_i)_d.$$

# Dimensions of Secants: State of the Art and Goal

## What is known:

- ▶ **Veronese:** Completely classified by the Alexander-Hirschowitz Theorem [AH95].
- ▶ **Grassmannians:** Many results, but open questions remain [BDG07].

## What is unknown:

- ▶ **Flag Varieties:** The landscape is much less understood.

## Main Goal of the Thesis

Extend the “Dual Terracini” dictionary to Flag Varieties using **Schur Apolarity**.

*Can we characterize the “apolar” space to the tangent space as the square of a suitable ideal? Can we use it to compute dimensions of secant varieties?*

$$(\hat{T}_P F)^\perp \stackrel{?}{=} I_{\mathcal{A}}^2(P)$$

# A Step Back: Schur Modules and Flag Varieties

**Irreducible representations of  $SL(V)$ :** *Schur modules*  $\mathbb{S}_\lambda V$ , indexed by partitions  $\lambda$ .

**Classical cases recover:**

- ▶  $\lambda = (d) \Rightarrow \mathbb{S}_\lambda V \simeq \text{Sym}^d V$  (symmetric tensors).
- ▶  $\lambda = (1^k) \Rightarrow \mathbb{S}_\lambda V \simeq \wedge^k V$  (skew-symmetric tensors).

**Flag varieties as homogeneous spaces:**

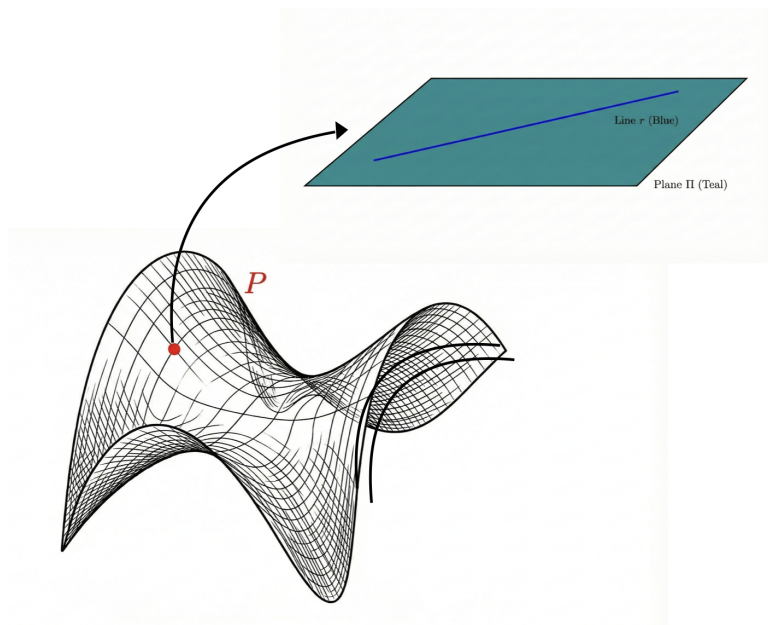
$$\mathbb{F}(n_1, \dots, n_k; V) \simeq SL(V)/P_{W_\bullet}.$$

The orbit of the highest weight vector in  $\mathbb{P}(\mathbb{S}_\lambda V)$  is a flag variety.

**Combinatorially:** For  $\lambda = (\lambda_1^{a_1}, \dots, \lambda_k^{a_k})$ , compute  $\lambda' = (n_k^{d_k}, \dots, n_1^{d_1})$ :

$$F := \mathbb{F}(n_1, \dots, n_k; V) \xhookrightarrow{\mathcal{O}(d_1, \dots, d_k)} \mathbb{P}(\mathbb{S}_\lambda V).$$

# A Step Back: Schur Modules and Flag Varieties



# Schur Apolarity and Dual Terracini

We work with flag varieties embedded in **Schur modules**  $\mathbb{S}_\lambda V$ . Our ambient algebra is  $(\mathbb{S}^\bullet V^*, \mathcal{M}_\mu^{\lambda, \nu})$ . We need to give a meaning to the apolar  $(\cdot)^\perp$ .

## Schur Apolarity Action

We use the Schur apolarity action [Sta23]:

$$* : \mathbb{S}_\mu V^* \otimes \mathbb{S}_\lambda V \longrightarrow \mathbb{S}_{\lambda/\mu} V.$$

This generalizes classical symmetric and skew-symmetric apolarity.

**Example:**  $((x_1 \wedge x_2) \otimes x_2) * ((e_1 \wedge e_2) \otimes e_1) = 1 \cdot 0 = 0$ .

For a point  $P \in \mathbb{P}(\mathbb{S}_\lambda V)$ , we define the **Schur Apolar Ideal**:

$$I_S(P) := \{f \in \mathbb{S}^\bullet V^* \mid f * P = 0\}.$$

## Natural Question

Can we characterize  $(\widehat{T}_P F)^\perp$  as the square of this ideal?

# A Partial Result: The Grassmannian Case

We first investigated the Grassmannian  $\mathbb{G}(k, V)$  embedded in  $\mathbb{P}(\mathbb{S}_{(1^k)} V)$ .

## Theorem (Dual Terracini for Grassmannians)

For the Grassmannian, both the geometric and Schur-theoretic approaches coincide:

$$(\widehat{T}_P \mathbb{G}(k, V))^\perp = I_S^2(P)_{(1^k)}.$$

- ▶ The algebraic square of the Schur apolar ideal  $I_S(P)$  correctly describes the apolar space to the tangent space (i.e., first-order neighborhood).
- ▶ Does the same work for arbitrary flag varieties  $F \subset \mathbb{P}(\mathbb{S}_\lambda V)$ ?

## Naive Guess

For any partition  $\lambda$ ,  $(\widehat{T}_P F)^\perp = I_S^2(P)_\lambda$ .

# Counterexample (with $\lambda = (4, 2)$ , $n = 3$ )

Consider the flag variety  $\mathbb{F}(1, 2; \mathbb{C}^3) \xrightarrow{\mathcal{O}(2,2)} \mathbb{P}(\mathbb{S}_{(4,2)}\mathbb{C}^3)$ .

Take  $\tilde{x} = (x_1 \wedge x_2)^{\otimes 2} \otimes x_2^{\otimes 2} \in (\hat{T}_P F)^\perp$ .

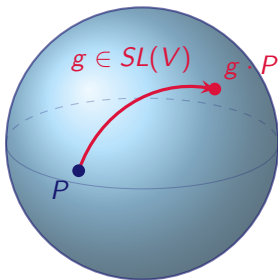
**Problem:** When trying to write  $\tilde{x}$  as a product in  $I_S^2(P)$ , we would need factors from  $I_S(P)$ . We showed that this **never happens**.

*Purely algebraic squares fail. Can we turn to a **more geometric** definition?*



**Key Idea:** We exploit the rational homogeneous structure of flag varieties!

# The Solution: A Geometric Approach



$$F \simeq SL(V)/\text{Stab}(P).$$

Infinitesimal action via Lie algebra.

We define the **Jet Ideal of order 2** (or Tangential Ideal) using derivations from  $\mathfrak{gl}(V)$ .

$$I^{(2)}(P) := \{f \in \mathbb{S}^\bullet V^* \mid f * P = 0 \text{ and } (\xi \cdot f) * P = 0, \forall \xi \in \mathfrak{gl}(V)\}.$$

# First Main Result: The Dual Terracini Lemma for Flags

Using this intrinsically defined ideal, we were able to prove the following.

## Theorem (Dual Terracini Lemma for Flags)

*For any flag variety embedded in a Schur module, we have*

$$(\widehat{T}_P F)^\perp = I^{\langle 2 \rangle}(P)_\lambda.$$

### Significance:

- ▶ This is the correct generalization of Lasker's result.
- ▶ It replaces the algebraic square  $I^2$  with the geometric  $I^{\langle 2 \rangle}$ .

**Something is still missing!** The ideal  $I^{\langle 2 \rangle}(P)_\lambda$  is somewhat obscure. Is there any way to handle this object more explicitly?

# Second Main Result: The Consistency Theorem

Schur module multiplication is expensive. Use an intermediate space!

$$F \xrightarrow{\Phi} \mathbb{P}\left(\bigotimes_{i=1}^k \operatorname{Sym}^{d_i}(\wedge^{n_i} V)\right) \xrightarrow{\bar{\pi}_\lambda} \mathbb{P}(\mathbb{S}_\lambda V).$$

**Slot-by-slot algebra:**

$$\mathcal{R}_\lambda := \bigotimes_{i=1}^k \operatorname{Sym}^\bullet(\wedge^{n_i} V^*).$$

**Slot ideal:** Generated by annihilators in each slot,

$$I_{\text{slot}}(P) = \left( (\wedge^{n_i} W_i)^\perp \text{ in slot } i \right).$$

## Consistency Theorem

If  $d_i \geq 2$  for all  $i$ :

$$\pi_\lambda(I_{\text{slot}}^2(P)_d) = I^{\langle 2 \rangle}(P)_\lambda,$$

where  $\pi_\lambda$  projects onto the Schur module.

# Application: Defectivity of Secant Varieties

**Example:** Take  $\lambda = (2^2)$ , corresponding to  $F = \mathbb{G}(2, \mathbb{C}^3)$  (Grassmannian), embedded with  $\mathcal{O}(2)$ .

Using Terracini + Schur duality:

$$\dim(\sigma_2(F)) = \dim(\mathbb{P}(\mathbb{S}_\lambda V)) - \dim((\widehat{T}_{P_1} F)^\perp \cap (\widehat{T}_{P_2} F)^\perp) - 1.$$

## Slot-by-slot computation:

- ▶ For generic points  $P_1, P_2$ , compute ideals in polynomial ring.
- ▶ Take intersection of their squares.
- ▶ Project back to dimension count.
- ▶ Find  $\dim(\sigma_2(F)) = 4 < 5 = \exp.\dim$ .

**Result:** Consistent with defectivity, done using polynomial algebra instead of direct geometry!

*This extends to any flag variety with a sufficiently ample embedding (and more!)*

# Computational Results (Secant Dimensions)

| Variety and embedding                                     | $r$ | Exp. dim | Found dim | Status               |
|-----------------------------------------------------------|-----|----------|-----------|----------------------|
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$       | 2   | 7        | 7         | regular              |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$       | 3   | 11       | 11        | regular              |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$       | 4   | 15       | 15        | regular              |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$       | 5   | 19       | 19        | regular              |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$       | 6   | 23       | 23        | regular              |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$       | 7   | 26       | 25        | defective (defect 1) |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(2, 2)$       | 8   | 26       | 26        | fills ambient space  |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(1, 1)$       | 2   | 7        | 6         | defective (defect 1) |
| $\mathbb{F}(1, 2; \mathbb{C}^3), \mathcal{O}(1, 1)$       | 3   | 7        | 7         | fills ambient space  |
| $\mathbb{F}(1, 2, 3; \mathbb{C}^4), \mathcal{O}(1, 1, 1)$ | 2   | 13       | 13        | regular              |
| $\mathbb{F}(1, 2, 3; \mathbb{C}^4), \mathcal{O}(1, 1, 1)$ | 3–7 | –        | –         | computing            |
| $\mathbb{G}(2, 3), \mathcal{O}(2)$                        | 2   | 5        | 4         | defective (defect 1) |
| $\mathbb{G}(2, 3), \mathcal{O}(2)$                        | 3   | 5        | 5         | fills ambient space  |

# Key Contributions & Future Work

## Contributions:

1. Proved Dual Terracini Lemma for Grassmannians in the Schur-theoretic setting.
2. Identified why naive algebraic approach fails for general partitions.
3. Proved Dual Terracini Lemma for flag varieties using jet ideal via Lie algebra derivations.
4. Established Consistency Theorem for slot-by-slot algebra.
5. Provided computational framework for secant dimensions.
6. Successfully run our first tests.

## Open Questions:

- ▶ Can we establish regularity results (classification) for secant varieties of flags using this method?
- ▶ What happens when embedding degrees  $d_i < 2$ ? Can we weaken the hypothesis of the Consistency Theorem?
- ▶ Which partitions allow Plücker relations to resolve ambiguities?

**Practical Impact:** *Shift complex Schur algebra computations to polynomial rings via slot-by-slot framework. Opens door to systematic computational study of flag variety geometry.*

**Thank you for your attention!**